

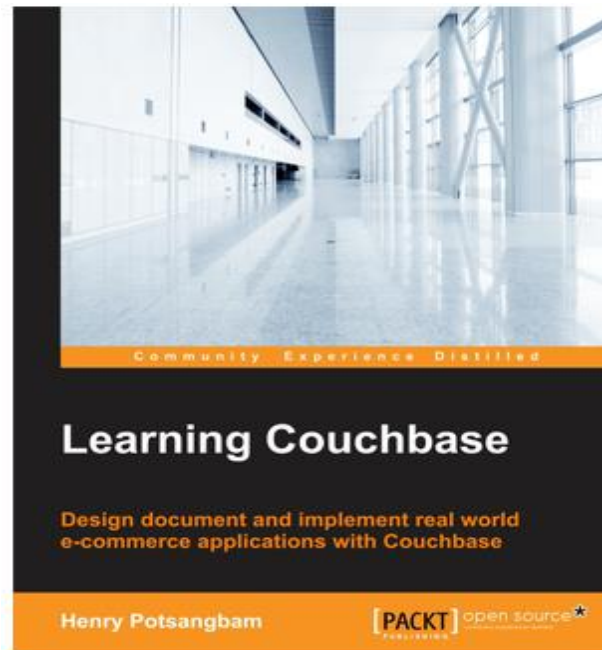
Kafka – Admins & Operations

Henry P

- **Certified Cassandra Admin**
- **Mapr Certified – Hadoop Administrator**
- **IBM Certified Application Developer**
- **IBM Certified Solution Designer**
- **SAP Certified ABAP & Portal Consultant .**
- **CIPM – Certificate in Project Management.**
- **TOGAF – Enterprise Architect**

IT Architect & Corporate trainer
20 +Year of IT Experience

NOSQL, Streaming Platform & Bigdata



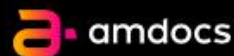
Author

Clientele

Tos



THOMSON REUTERS



Day	Topic	Key Concepts	Hands-on Lab
Day 1	Introduction to Kafka and Setup	- Overview of Kafka - Kafka Architecture (Zookeeper, KRaft, Brokers, Topics, Partitions) - Core Concepts: Producers, Consumers, Clusters	- Creating and managing Kafka topics - Configuring partitions, replication, retention
Day 2	Kafka Administration Basics	- Kafka Ecosystem Components ; Control Center - Kafka Schema Management (Avro, Schema Registry) - Scaling Kafka Clusters	- Configuring consumer groups and offsets - Producer configurations
Day 3	Kafka Operations	- Operations & Maintenance: Backup, Recovery, Upgrades, Troubleshooting , Security - Cluster Management: Adding/Removing Brokers - Log Management	- Configuring retention and log compaction - Managing partitions, adding replicas
Day 4	Kafka Advanced Features and Optimization	- Kafka Connect: Custom Connectors, Troubleshooting - Multi-Datcenter Replication (MirrorMaker) - Disaster Recovery	- Integrating telemetry - Implementing performance tuning - Troubleshooting Kafka performance
Day 5	Kafka Monitoring & Troubleshooting	- Kafka Performance Tuning , Kubernetes Deployment, Advance API [Kstream + KSQL] - Kafka Monitoring (Metrics, JMX, Alerts) - Telemetry (OpenTelemetry, Prometheus, Grafana)	- Deploying Kafka in Kubernetes using Strimzi

Introduce Yourself.

Name

Year of Experience.

Skills Level

Java / Linux

Messaging System / Kafka

Expectation, if any.

Note: Basic knowledge of Java & Linux are required.

Schedule

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Time	
9.30 – 11.00 AM	Session I
11.00 AM to 11.15 AM	Tea Break
11.15 AM to 12.45 PM	Session II
12.45 PM to 1.45 PM	Lunch Break
1.45 PM to 3.15 PM	Session III
3.15 PM to 3.30 PM	Tea Break
3.30 PM to 5.30 PM	Session IV



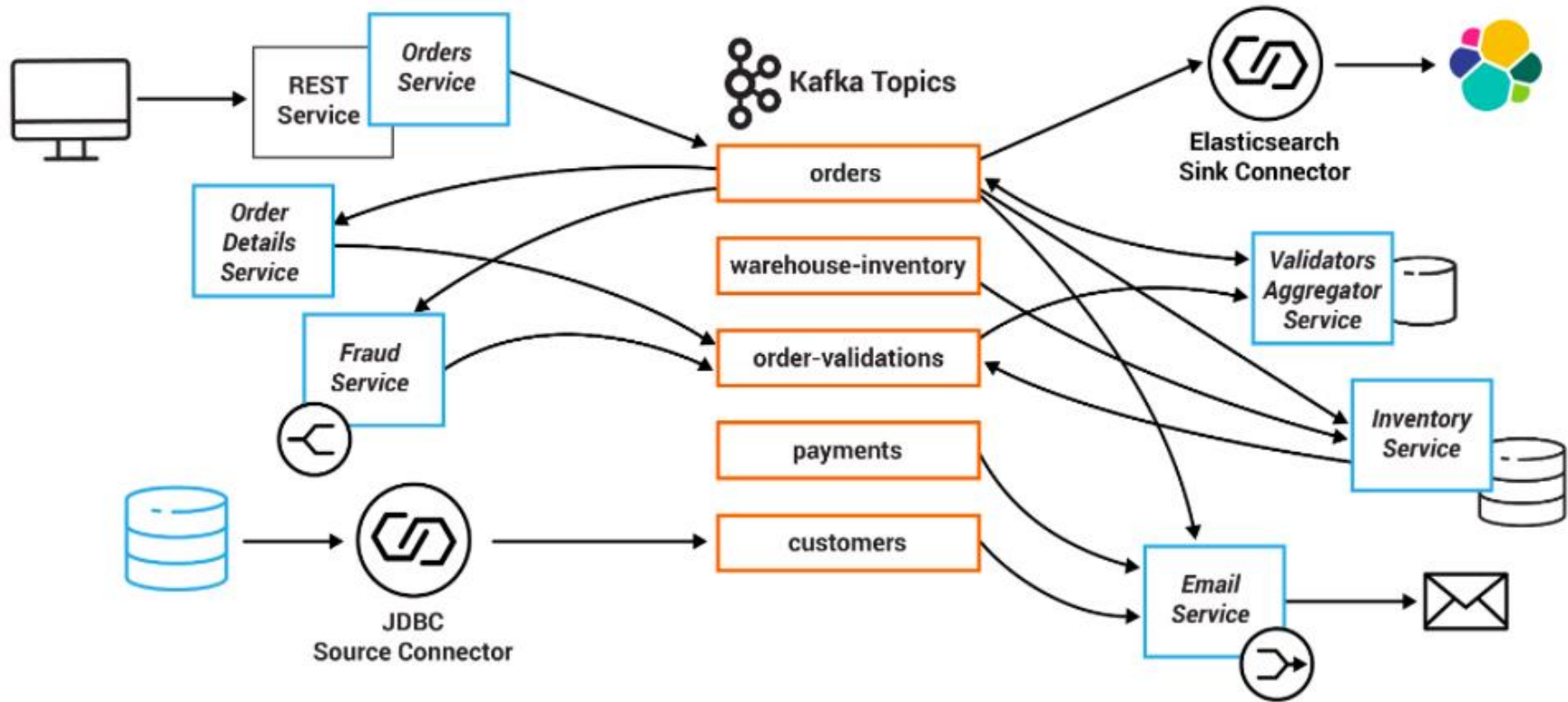
16 March 2025

Kafka – An Overview

- ❖ 1/3 of all Fortune 500 companies
- ❖ Top ten travel companies, 7 of ten top banks, 8 of ten top insurance companies, 9 of ten top telecom companies
- ❖ LinkedIn, Microsoft and Netflix process 4 comma message a day with Kafka (1,000,000,000,000)
- ❖ Real-time streams of data, used to collect big data or to do real time analysis (or both)

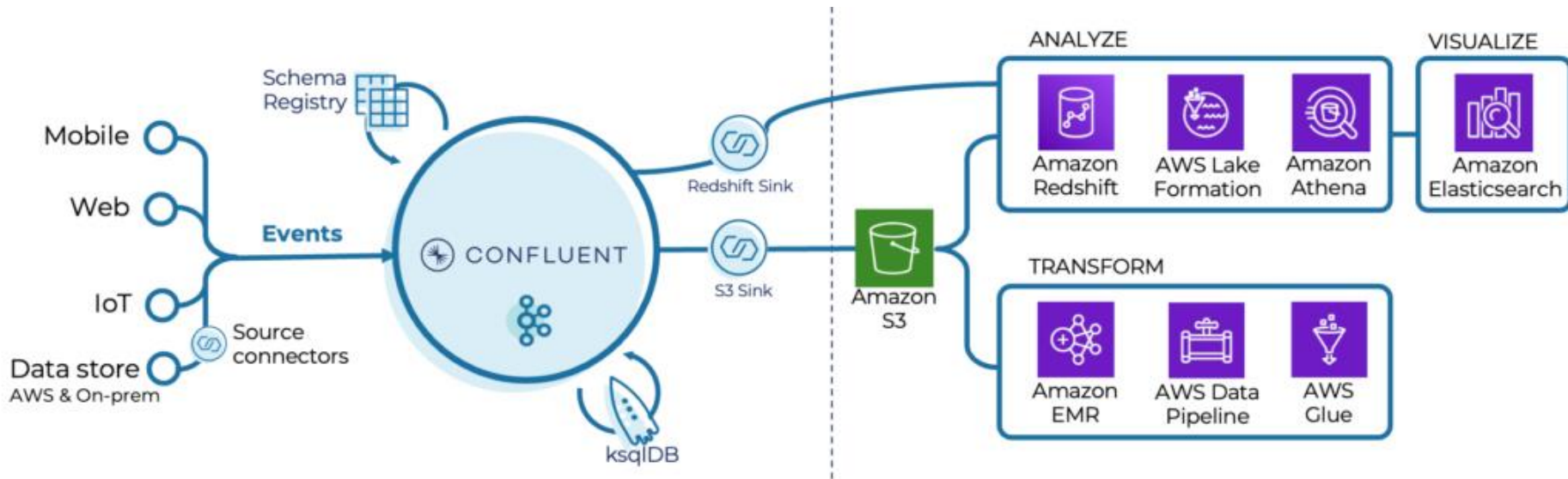
Why Kafka is Needed? - MicroService

Tos



Why Kafka is Needed? – Big Data

Tos



Why Kafka is Needed?

- ❖ Apache Kafka is a fast, scalable, durable, and fault- tolerant publish-subscribe messaging system
- ❖ Real time streaming data processed for real time analytics
- ❖ Service calls, track every call, IOT sensors
- ❖ Kafka is often used instead of JMS, RabbitMQ and AMQP
- ❖ higher throughput, reliability and replication

- ❖ Kafka can work in combination with
 - Flume/Flafka, Spark Streaming, Storm, HBase and Spark for real-time analysis and processing of streaming data
 - Feed your data lakes with data streams
- ❖ Kafka brokers support massive message streams for follow-up analysis in Hadoop or Spark
- ❖ Kafka Streaming (subproject) can be used for real-time analytics

- ❖ Build real-time streaming applications that react to streams
 - ❖ Real-time data analytics
 - ❖ Transform, react, aggregate, join real-time data flows
 - ❖ Feed events to CEP for complex event processing
 - ❖ Feed data lakes
- ❖ Build real-time streaming data pipe-lines
 - ❖ Enable in-memory micro services (actors, Akka, Vert.x, Qbit, RxJava)

Why is Kafka Popular?

- ❖ ***Great performance***
- ❖ Operational Simplicity, easy to setup and use, easy to reason
- ❖ Stable, Reliable Durability,
- ❖ Flexible Publish-subscribe/queue (scales with N-number of consumer groups),
- ❖ Robust Replication,
- ❖ Works well with systems that have data streams to process, aggregate, transform & load into other stores

Why is Kafka so fast?

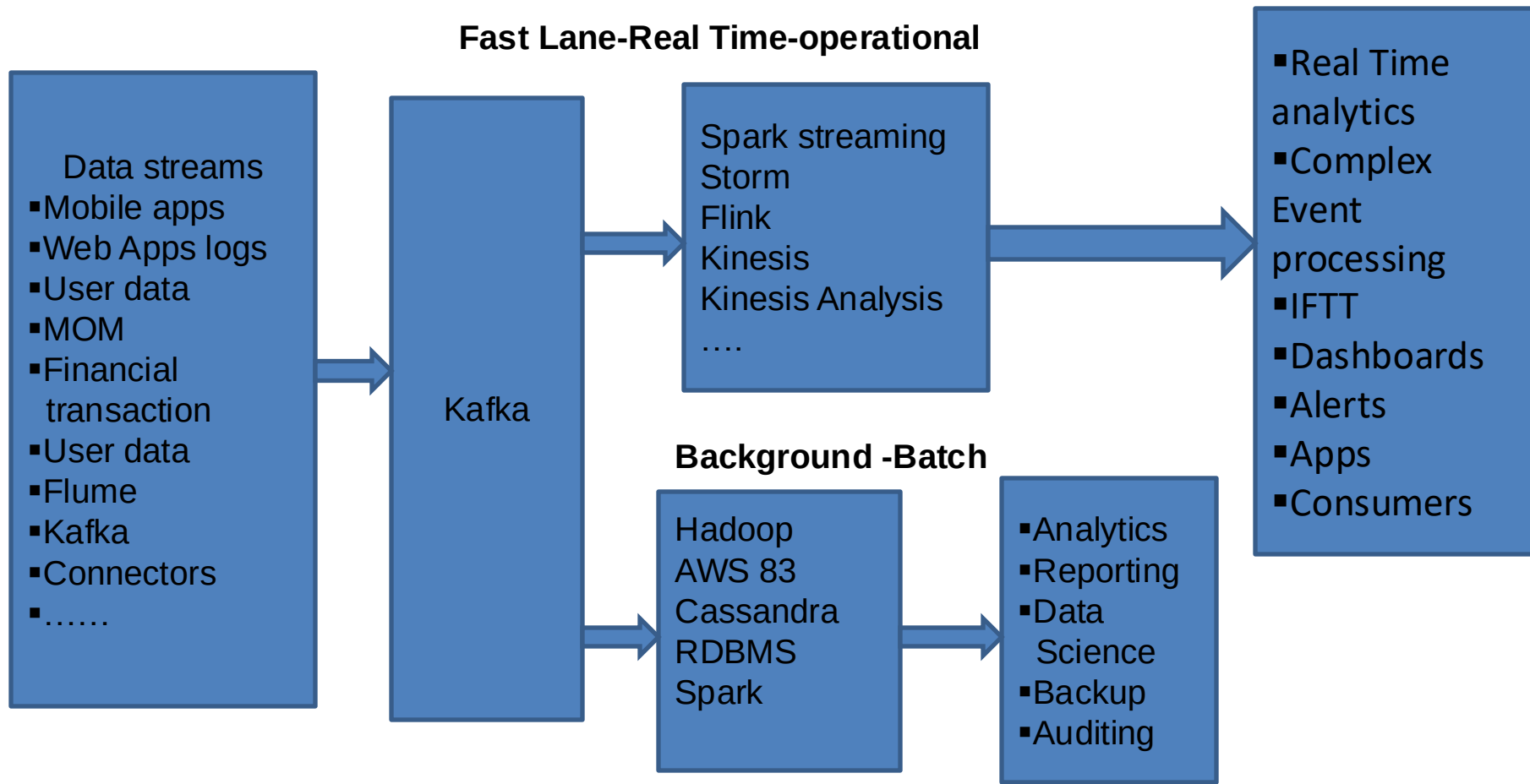
- ❖ **Zero Copy** - calls the OS kernel direct rather to move data fast
- ❖ **Batch Data in Chunks** - Batches data into chunks
 - ❖ end to end from Producer to file system to Consumer
 - ❖ Provides More efficient data compression. Reduces I/O latency
- ❖ **Sequential Disk Writes** - Avoids Random Disk Access
 - ❖ writes to immutable commit log. No slow disk seeking. No random I/O operations. Disk accessed in sequential manner
- ❖ **Horizontal Scale** - uses 100s to thousands of partitions for a single topic
 - ❖ spread out to thousands of servers
 - ❖ handle massive load

Kafka: A Stream Data Platform

Kafka Streaming Architecture

Tos

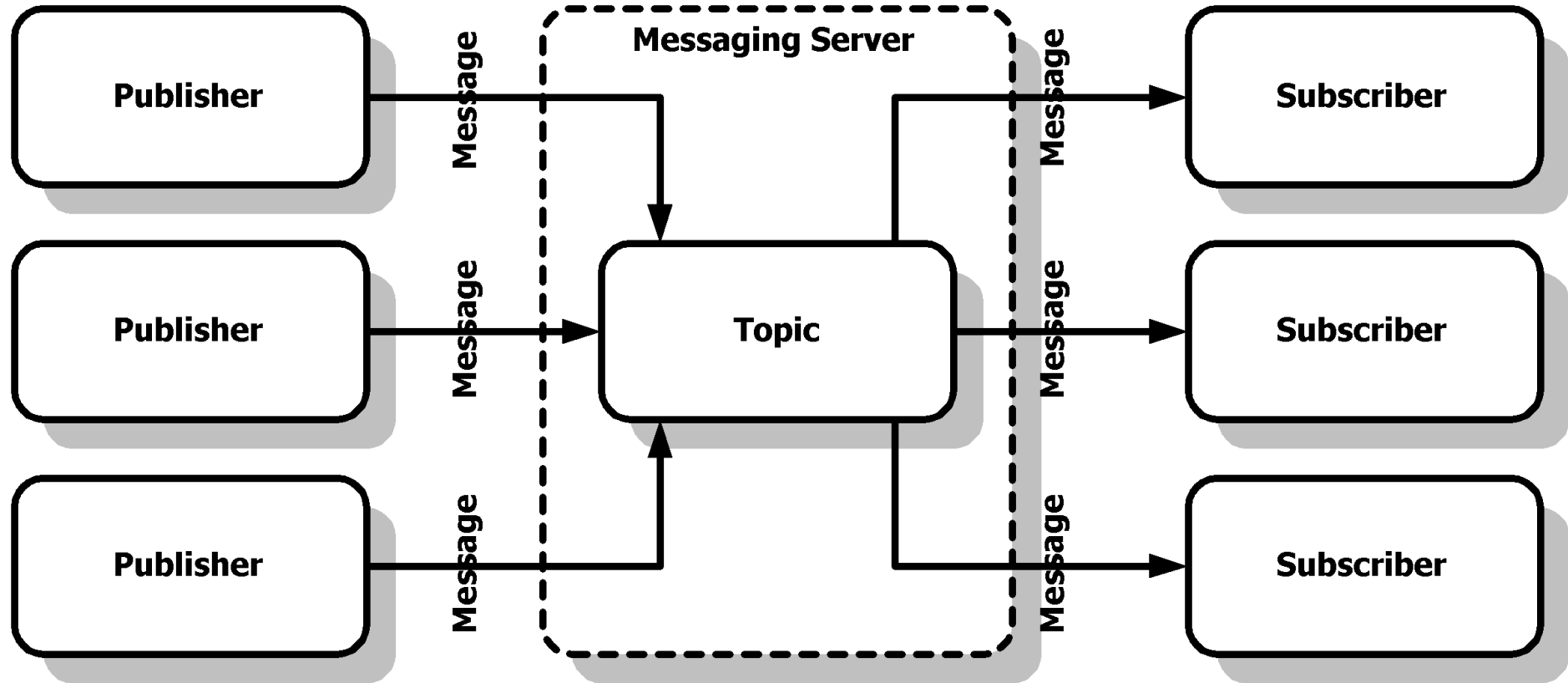
Fast Lane-Real Time-operational



What is Kafka?

What is Kafka?

Tos



What is Kafka?

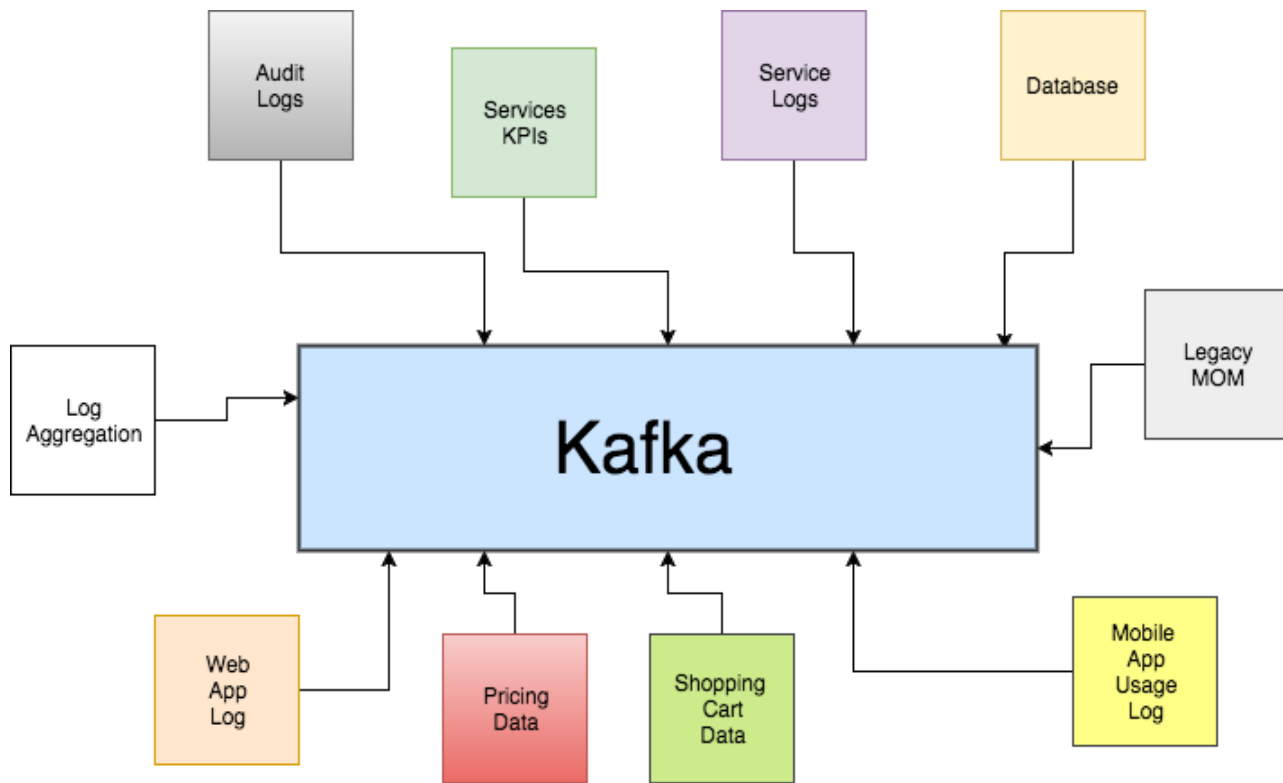
- ❖ Distributed Streaming Platform
 - ❖ Publish and Subscribe to streams of records
 - ❖ Fault tolerant storage
 - ❖ Replicates Topic Log Partitions to multiple servers
 - ❖ Process records as they occur
 - ❖ Fast, efficient IO, batching, compression, and more
- ❖ Used to decouple data streams

- ❖ Kafka decouple data streams
- ❖ producers don't know about consumers
- ❖ Flexible message consumption
 - ❖ Kafka broker delegates log partition offset (location) to Consumers (clients)

- ❖ Feeding of high-latency daily or hourly data analysis into Spark, Hadoop, etc.
- ❖ Feeding micro services real-time messages
- ❖ Sending events to CEP system
- ❖ Feeding data to do real-time analytic systems
- ❖ Up to date dashboards and summaries
- ❖ At same time

Kafka Decoupling Data Streams

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Don't couple the stream

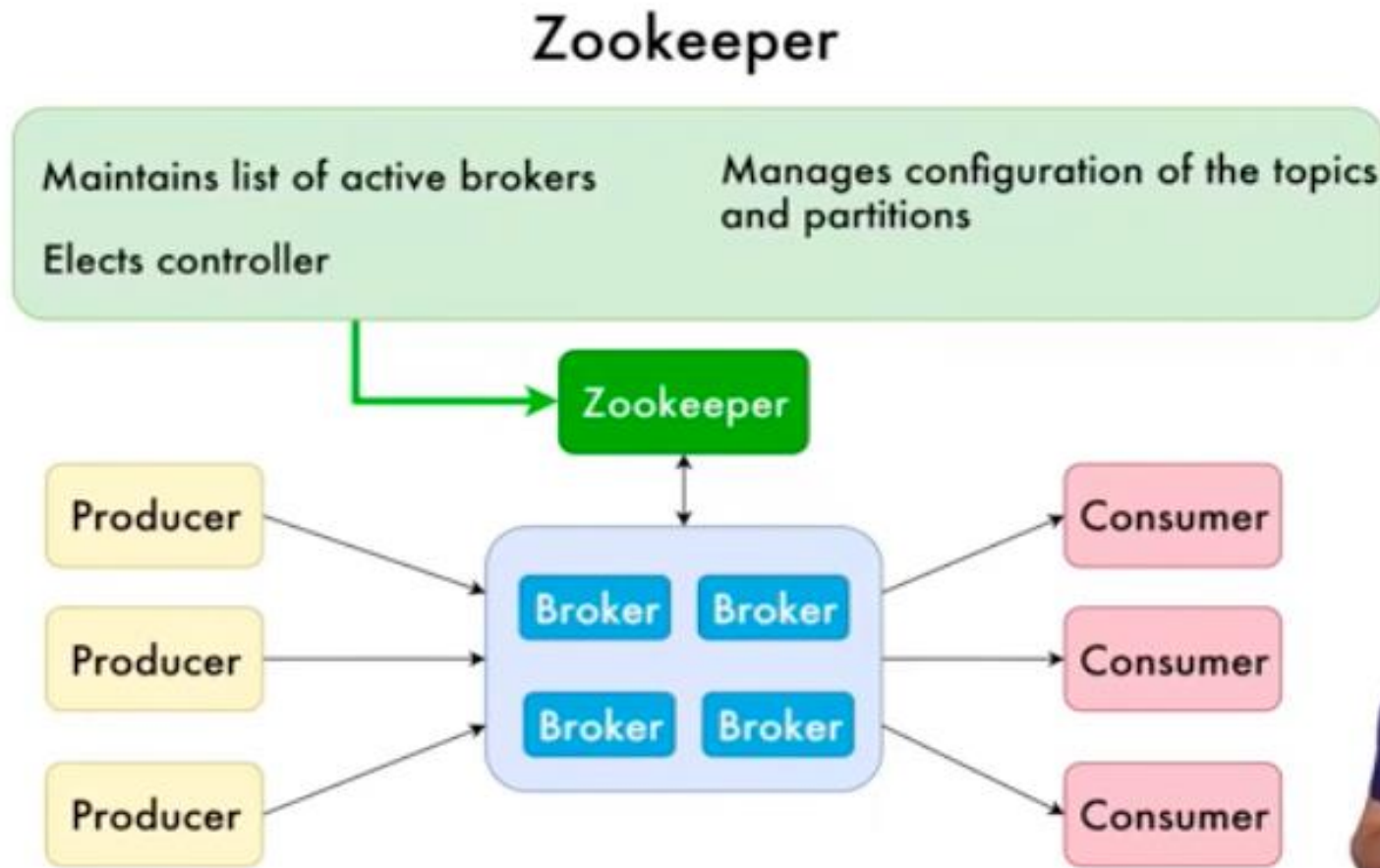


- ❖ Kafka communication from clients and servers wire protocol over TCP protocol
- ❖ Protocol versioned
- ❖ Maintains backwards compatibility
- ❖ Many languages supported
- ❖ Kafka REST proxy allows easy integration (not part of core)
- ❖ Also provides Avro/Schema registry support via Kafka ecosystem (not part of core)

Kafka Architecture

Kafka: Topics, Producers, and Consumers

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- ❖ **Broker:** Kafka server that runs in a Kafka Cluster. Brokers form a cluster. Cluster consists on many Kafka Brokers on many servers.
- ❖ **Zoo Keeper:** Does coordination of brokers/cluster topology. Consistent file system for configuration information and leadership election for Broker Topic Partition Leaders

Installation steps

Installing Java

```
#tar -xvf jdk-8u45-linux-x64.tar.gz -C /opt
```

Set in the path variable and JAVA_HOME

Include in the profile a follow

```
[root@tos opt]# more ~/.bashrc
# .bashrc

# User specific aliases and functions

alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'

export JAVA_HOME=/opt/jdk1.8.0_45

export PATH=$PATH:$JAVA_HOME/bin
# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi
[root@tos opt]#
```

Installation steps...

Installing Zookeeper

The following example installs Zookeeper with a basic configuration in `/opt/zookeeper`, storing its data in `/opt/data/zookeeper`:

```
# tar -xvf zookeeper-3.4.12.tar.gz -C /opt
```

```
# mv zookeeper-3.4.12 /opt/zookeeper
```

```
# mkdir -p /opt/data/zookeeper
```

```
# vi /opt/zookeeper/conf/zoo.cfg
```

```
tickTime=2000
```

```
dataDir=/opt/data/zookeeper
```

```
clientPort=2181
```

```
# /opt/zookeeper/bin/zkServer.sh start
```

```
[root@tos opt]# /opt/zookeeper/bin/zkServer.sh start
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
[root@tos opt]#
```

Installation steps..

Installing a Kafka Broker

The following example installs Kafka in `/opt/kafka`, configured to use the

Zookeeper server started previously and to store the message log segments stored in `/tmp/kafka-logs`:

```
# tar -zxf kafka_2.12-1.1.0.tgz -C /opt
# mv kafka_2.12-1.1.0 /opt/kafka
# mkdir /opt/data/kafka-logs
# /opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/server.properties
```

```
[root@tos opt]# /opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/s
erver.properties
[root@tos opt]# jps
3476 Kafka
3499 Jps
2895 QuorumPeerMain
[root@tos opt]#
```

Lab : Installation of Kafka